Politecnico di Milano

Dipartimento di Elettronica, Informazione e Bioingegneria

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Advanced Algorithms and Parallel Programming – II part – February 14th, 2023

Polimi ID _____ _____

Surname _____ _____ **Name** _____

- This is a closed-book examination. You cannot use computers, phones, or laptops during the exam.
- Paper will be provided, but you should bring and use writing instruments that yield marks dark enough to be read easily. Erasable pens can be used.
- Total available time: 1h:15m.

Exercise 1 (4 points) _____

Exercise 2 (4 points) _____

Exercise 3 (4 points) _____

Exercise 4 (4 points) _____

Exercise n. 1

Answer the following questions about Parallel Algorithms and Parallel Programming and briefly explain (without an explanation, the answer will be considered invalid)

A. "Parallel execution must address control, data, and system dependences". What is meant by "dependency" in this context? (1)

B. Is the following code parallelizable? Are there better parallelizable solutions? (2)

```
for (j=1; j<100; j++)
        for (i=0; i<100; i++)
            a[i][j] = f(a[i][j-1]);
```

C. Please explain the differences between "*loop-carried*" and "*loop-independent".* (1)

Exercise n. 2

Answer the following questions about parallel patterns and briefly explain (without an explanation, the answer will be considered invalid)

A. Please describe the map pattern: assumptions, complexity, example. (1)

B. Please describe the differences between Zip and Unzip parallel patterns. (1)

C. Describe all the collision rules for a scatter pattern. (1)

D. What is meant by "*permutation scatter*"? (1)

Exercise n. 3

Answer the following questions about OpenMP and briefly explain (without an explanation, the answer will be considered invalid)

A. Is it possible to implement a pipeline paradigm using OpenMP? What parallel paradigm is adopted by default in OpenMP?(2)

B. Is it possible to control the visibility of variables between different threads in OpenMP? In which ways? (2)

Exercise n. 4

Answer the following questions about MPI and PThreads and briefly explain (without explanation the answer will be considered invalid)

A. Describe the functionality of the MPI_Reduce subroutine (You can skip this question if you completed the second challenge) (2)

B. What is the difference between joinable and detached threads in PThread? Can detached threads' execution be synchronized with other threads? (2)